



HVDC Load Flow

DC power flow, current and losses for symmetrical monopole and bipole configurations

Revision history

REV.	DATE	AUTHOR	DESCRIPTION
0.1	Sep. 8, 2026	Eugen Starschich	Initial document
0.2	Sep. 8, 2026	Eugen Starschich	Conductor temperature correction; four case columns; one-pole-out case specified by power
0.3	Sep. 8, 2026	Eugen Starschich	Reference case aligned with current design inputs (Chukar DMR)
0.4	Sep. 9, 2026	Eugen Starschich	Refocused on the physical description; implementation section removed
0.5	Sep. 10, 2026	Eugen Starschich	HVDC element notation applied (U_{dpp} , U_{dpe} , I_d)
0.6	Sep. 10, 2026	Eugen Starschich	DC-side notation figure added; voltages qualified by station
0.7	Sep. 10, 2026	Eugen Starschich	Bipole notation figure added; resistances and currents renamed to $R_{d,hv}$, $R_{d,dmr}$, $I_{d,dmr}$
0.8	Sep. 10, 2026	Eugen Starschich	Conductor table reference added; data basis corrected to DC at 20 °C and all results recomputed
0.9	Sep. 10, 2026	Eugen Starschich	References moved from footnotes to a closing References section

CONTENT

1	Scope and conventions	4
1.1	Purpose	4
1.2	Direction and reference conventions	4
1.3	Conductor data	4
1.4	Notation	4
2	Configurations covered	7
2.1	Symmetrical monopole	7
2.2	Bipole with dedicated metallic return	7
2.3	Reduced-capacity operating state	7
3	Model	9
3.1	Conductor resistance and temperature	9
3.2	Line resistance	9
3.3	Converter stations	10
3.4	DC line	10
3.5	Receiving-end voltage	10
4	Solution of the load flow	11
4.1	Why the current is not explicit	11
4.2	The current	11
4.3	Maximum transferable power	11
4.4	The rest of the flow	12
4.5	Current margin	12
5	Worked cases	13
5.1	The cases	13
5.2	Temperature correction and line resistance	13
5.3	Balanced bipole, step by step	14
5.4	One pole out, step by step	14
5.5	Results	14
5.6	Reading the results	14
6	Assumptions and limitations	16
6.1	Assumptions	16
6.2	Outside the scope	16
6.3	Open items	16
7	References	18

1 Scope and conventions

1.1 Purpose

This document describes the steady-state power flow of a point-to-point HVDC link: Assuming the power to be defined at the AC terminal of the receiving end. Two HVDC topologies arrangements are covered — the **symmetrical monopole (SMP)** and the **bipole with a dedicated metallic return (BIP)**.

1.2 Direction and reference conventions

Power flows from the **rectifier** (sending station) to the **inverter** (receiving station). Four conventions fix the problem:

- (1) **Power is defined at the receiving end**, at the AC terminals of the inverter station.
- (2) **DC voltage is regulated at the rectifier DC terminals**. The nominal dc voltage $U_{dpe,rec}$ is defined at the sending station and the receiving-end voltage comes out lower by the resistive drop of the line.
- (3) **Converter station losses are a fraction of station throughput**, applied per station, rather than a fixed number of megawatts.
- (4) **Conductor resistance is corrected to a design temperature**. Conductor data is quoted at 20 °C; the flow is evaluated at a conductor temperature stated separately, 75 °C by default.

Conventions (1) and (2) sit at opposite ends of the line, which is what makes the current an implicit rather than an explicit quantity: the sending-end power depends on the line loss, the line loss depends on the current, and the current depends on the sending-end power. Section 4 closes that loop in one step instead of iterating.

1.3 Conductor data

The one genuine conversion concerns the conductor data. Conductor tables give DC resistance per 1000 ft — the values used here are the *DC @ 20 °C* column of the Southwire ACSR bare conductor tables [1] — so the factor

$$\frac{5280 \text{ ft/mi}}{1000 \text{ ft}} = 5.28$$

converts specific resistance to Ω/mi before multiplying by the line length in miles.

1.4 Notation

Figures 1 and 2 show the signal notation on the DC side, for the SMP and BIP topologies.

Each quantity carries the station it is measured at, *rec* or *inv*. In the bipole it carries the pole as well, *pos* or *neg*, since each pole has its own converter pair; the HV conductor and the DMR are distinguished in both the resistances, $R_{d_{hv}}$ and $R_{d_{dmr}}$, and the currents, $I_{d_{hv}}$ and $I_{d_{dmr}}$; and U_{dmr} is the DMR potential at the station where the return is not earthed. The relations in the sections that follow use the unqualified symbol wherever a statement holds for either pole or for either station.

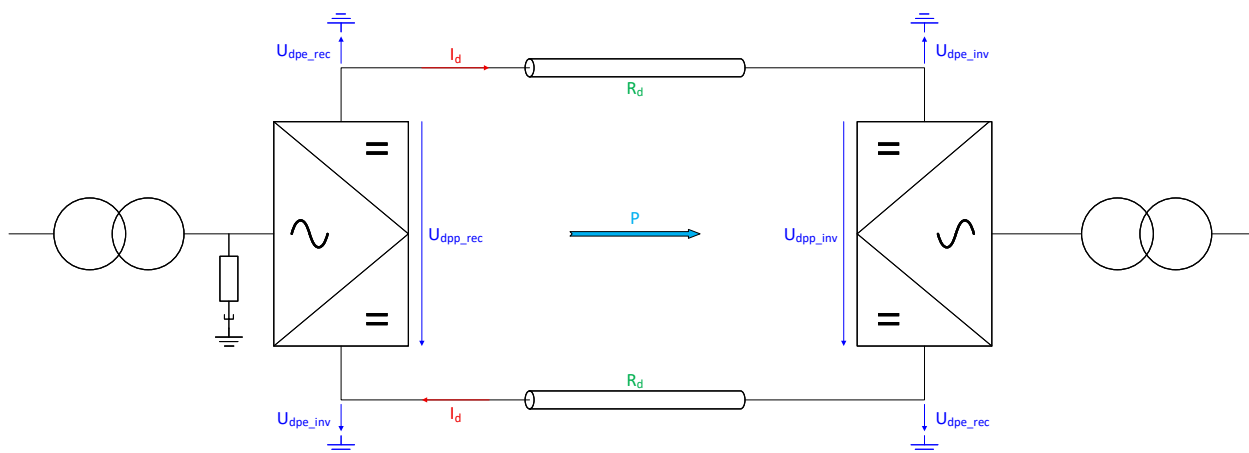


Figure 1: DC-side notation for SMP

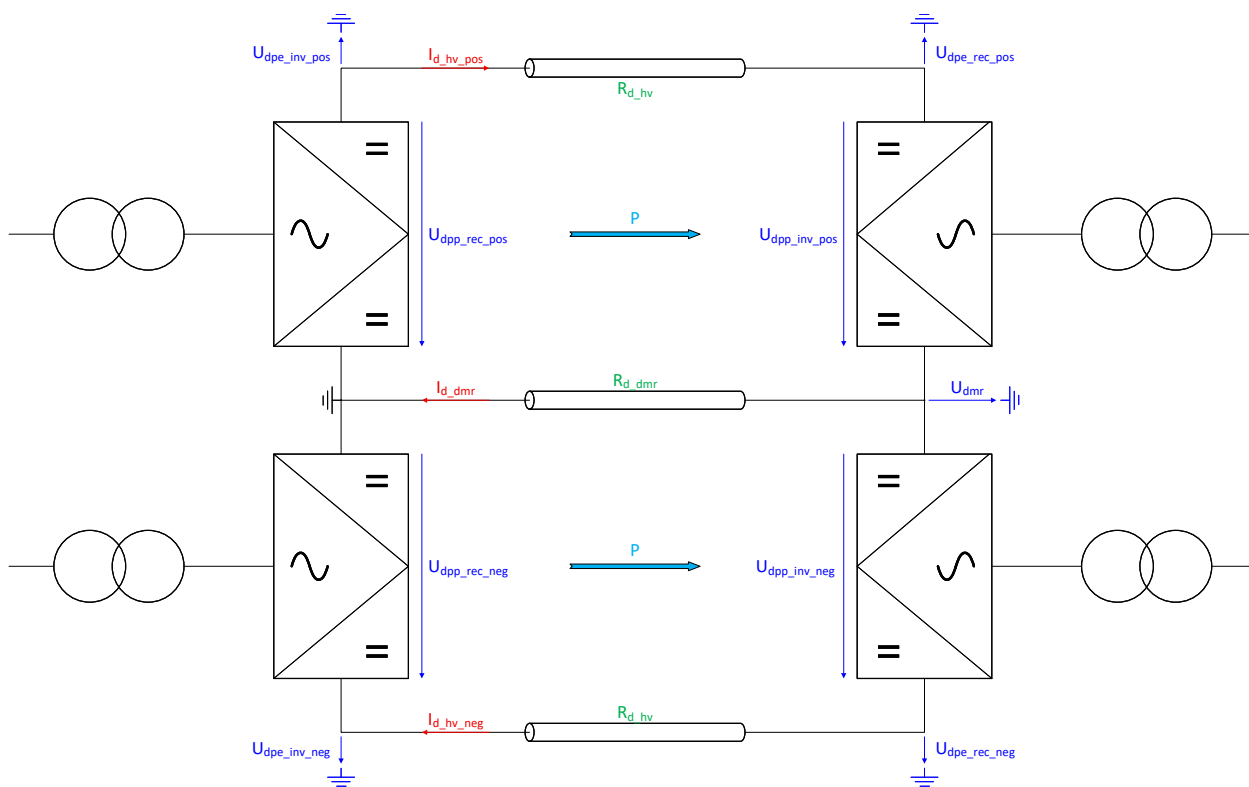


Figure 2: DC-side notation for BIP

Symbol	Unit	Quantity
$P_{ac,inv}$	MW	AC power at the receiving end (inverter AC terminals) — <i>given</i>
$P_{ac,rec}$	MW	AC power drawn at the sending end (rectifier AC terminals)
$P_{dc,inv}$	MW	DC power at the inverter DC terminals
$P_{dc,rec}$	MW	DC power at the rectifier DC terminals
U_{dpe}	kV	DC pole-to-ground (earth) voltage
$U_{dpe,rec}$	kV	DC pole-to-ground voltage at the rectifier — <i>given</i>
$U_{dpe,inv}$	kV	DC pole-to-ground voltage at the inverter
U_{dpp}	kV	DC voltage across a converter's DC terminals; $U_{dpp} = 2 \cdot U_{dpe}$ for a symmetrical monopole
U_{dmr}	kV	DMR potential at the station where the return is not earthed
$U_{dpm,inv}$	kV	DC pole-to-DMR voltage at the inverter (one pole out)
I_d	kA	DC pole current, operating value
I_{rated}	kA	DC pole current including design margin
k	—	Number of energised poles: 2 normally, 1 with a pole out
L	mi	DC line length
$r'_{d,hv}$	$\Omega/1000$ ft	Specific resistance of one HV subconductor
n_{hv}	—	Subconductors per pole (bundle count)
$R_{d,hv}$	Ω	Resistance of one pole conductor over the whole length
$r'_{d,dmr}$	$\Omega/1000$ ft	Specific resistance of one DMR subconductor
n_{dmr}	—	Subconductors in the DMR bundle
$R_{d,dmr}$	Ω	Resistance of the DMR over the whole length
$I_{d,dmr}$	kA	Current in the DMR; zero when the bipole is balanced
R_{loop}	Ω	Resistance of the complete current path (out and return)
T_{ref}	$^{\circ}\text{C}$	Temperature the conductor resistance data refers to
T	$^{\circ}\text{C}$	Conductor temperature the flow is evaluated at
T_0	$^{\circ}\text{C}$	Inferred zero-resistance temperature of the conductor material
k_T	—	Temperature correction factor on the specific resistance
p_{rec}	—	Rectifier station loss fraction of throughput
p_{inv}	—	Inverter station loss fraction of throughput
m	—	Margin applied to the DC current for rating purposes
$P_{loss,line}$	MW	Total resistive loss of the DC line
η	—	Overall AC-to-AC efficiency of the link

Table 1: Notation.

2 Configurations covered

2.1 Symmetrical monopole

The symmetrical monopole carries the full power on two conductors held at $+U_{dpe}$ and $-U_{dpe}$ against ground, with no third conductor and no ground path in service. The DC circuit is a single loop: current leaves the rectifier on the positive conductor and returns on the negative one. The voltage seen by the converter is therefore the pole-to-pole voltage

$$U_{dpp} = 2 \cdot U_{dpe},$$

and the transmitted power is

$$P_{dc} = U_{dpp} \cdot I_d.$$

Because there is no return path other than the second conductor, the loss of either conductor is the loss of the whole link — there is no partial operating mode, which is the reason a bipole is preferred where availability matters.

2.2 Bipole with dedicated metallic return

The bipole runs two independently controlled poles at $\pm U_{dpe}$ plus a third, lower-rated conductor: the dedicated metallic return. In normal **balanced** operation the two pole currents are equal and opposite, the DMR carries no current, and the circuit is electrically identical to the symmetrical monopole:

$$P_{dc} = 2 \cdot U_{dpe} \cdot I_d, \quad I_{d,dmr} = 0.$$

The DMR earns its place when the bipole becomes unbalanced. With one pole out of service the remaining pole keeps operating and its current returns through the DMR instead of the lost conductor, so the link continues at reduced capacity. In that condition the circuit is a single-pole loop:

$$P_{dc} = U_{dpe} \cdot I_d,$$

and the return path resistance is that of the DMR rather than that of a second HV bundle — which is why the DMR conductor is a separate input in the calculation, as it is generally not the same conductor as the poles.

2.3 Reduced-capacity operating state

With one pole out of service the link is a different circuit, and it is treated as its own operating state rather than as a fraction of the healthy one: a post-contingency transfer is set at the receiving end and the current that delivers it through one pole and the DMR follows. The plant is unchanged — same voltage, length, conductors, station losses and conductor temperature — so only the power and the return path distinguish the state from balanced operation.

Two consequences are worth stating. The pole current in this state is *not* half the bipolar current: at half the power on one pole instead of two, the current is of the same order as the full bipolar value, and with the more resistive DMR in the return path it comes out slightly above it. It is therefore the outage state, not normal operation, that can set the conductor and valve rating. And because the same order of current flows through a loop of higher resistance while carrying half the power, the loss as a fraction of the transfer is several times worse than in balanced operation.

Operating state	P_{dc}	R_{loop}	Current path
Symmetrical monopole	$U_{dpp} \cdot I_d$	$2 \cdot R_{d,hv}$	Out on the positive conductor, back on the negative conductor.
Bipole, balanced	$2 \cdot U_{dpe} \cdot I_d$	$2 \cdot R_{d,hv}$	Out on the positive pole, back on the negative pole; DMR idle.
Bipole, one pole out	$U_{dpe} \cdot I_d$	$R_{d,hv} + R_{d,dmr}$	Out on the healthy pole, back through the DMR.

Table 2: The three operating states covered. The first two obey the same equations; only the third involves the DMR.

3 Model

The link is represented by three elements in series: the rectifier station, the DC line, and the inverter station. Each is a single algebraic relation.

3.1 Conductor resistance and temperature

Conductor tables quote DC resistance per unit length at a stated reference temperature. The data used here is given at $T_{ref} = 20^\circ\text{C}$, while the line in service runs hot, so the specific resistance is corrected to the design conductor temperature T before anything else is calculated.

The correction uses the inferred zero-resistance temperature T_0 — the temperature at which the linear resistance-temperature characteristic of the metal extrapolates to zero:

$$k_T = \frac{T_0 + T}{T_0 + T_{ref}}, \quad r'(T) = r'(T_{ref}) \cdot k_T. \quad (1)$$

For hard-drawn 1350-H19 aluminium, which carries essentially all of the current in an ACSR conductor, $T_0 = 228^\circ\text{C}$. This is the same statement as a temperature coefficient $\alpha_{20} = 1/(228 + 20) = 0.00403\text{ K}^{-1}$ referred to 20°C , written in the form that avoids having to restate α every time the reference temperature changes. The constant is a property of the material, so a DMR of a different alloy is corrected with its own value.

With the default design temperature of 75°C — the usual maximum continuous operating temperature for an ACSR conductor — the factor is

$$k_T = \frac{228 + 75}{228 + 20} = \frac{303}{248} = 1.2218, \quad (2)$$

that is, the line is some 22 % more resistive hot than the conductor table suggests. Since line losses scale with resistance, ignoring the correction would understate them by the same 22 %, which is why the temperature basis belongs in the calculation rather than in a footnote.

The correction can be checked against the conductor table itself, which also lists an AC resistance at 75°C . For Bluebird, $0.00801 \cdot 1.2218 = 0.00979\ \Omega/1000\text{ ft}$ against a tabulated $0.0105\ \Omega/1000\text{ ft}$; the remaining 7 % is skin effect, which is why the tabulated AC value is not the one to use on a DC line even though its temperature is right.

3.2 Line resistance

The resistance of one pole conductor over the whole route is the corrected specific resistance of a single subconductor, divided by the number of subconductors in the bundle (they are electrically in parallel), converted to Ω/mi and multiplied by the length:

$$R_{d,hv} = \frac{r'_{d,hv} \cdot 5.28}{n_{hv}} \cdot L \cdot k_T, \quad R_{d,dmr} = \frac{r'_{d,dmr} \cdot 5.28}{n_{dmr}} \cdot L \cdot k_T. \quad (3)$$

The resistance that matters for the flow is that of the complete circuit — out and back — which depends on the configuration:

$$R_{loop} = \begin{cases} 2 \cdot R_{d,hv} & \text{symmetrical monopole, and balanced bipole,} \\ R_{d,hv} + R_{d,dmr} & \text{bipole with one pole out (DMR return).} \end{cases} \quad (4)$$

Equation (3) is a DC resistance. No skin or proximity effect is applied; on a DC line neither arises, and temperature is the one correction that matters.

3.3 Converter stations

Station losses are taken as a fraction of the power passing through the station. The two stations therefore differ only in which side of the conversion is known:

$$\text{rectifier (AC} \rightarrow \text{DC): } P_{dc,rec} = P_{ac,rec} \cdot (1 - p_{rec}) \iff P_{ac,rec} = \frac{P_{dc,rec}}{1 - p_{rec}}, \quad (5)$$

$$\text{inverter (DC} \rightarrow \text{AC): } P_{ac,inv} = P_{dc,inv} \cdot (1 - p_{inv}) \iff P_{dc,inv} = \frac{P_{ac,inv}}{1 - p_{inv}}. \quad (6)$$

A single fraction lumps everything inside the station fence: valve conduction and switching losses, converter transformer and reactor losses, cooling and auxiliary supply. The loss fraction is a straight percentage of throughput, so it scales linearly with power and does not capture the no-load component of a real loss curve; see Section 6.

Because power is specified at the receiving AC terminals, Equation (6) is the *entry point* of the whole calculation: it converts the given $P_{ac,inv}$ into the DC power that must arrive at the inverter DC terminals.

3.4 DC line

With the DC voltage held at $U_{dpe,rec}$ pole-to-ground at the rectifier, the power leaving the rectifier and the resistive loss of the line are

$$P_{dc,rec} = k \cdot U_{dpe,rec} \cdot I_d, \quad P_{loss,line} = R_{loop} \cdot I_d^2, \quad k = \begin{cases} 2 & \text{two poles energised,} \\ 1 & \text{one pole out,} \end{cases} \quad (7)$$

and the power arriving at the inverter DC terminals is the difference:

$$P_{dc,inv} = k \cdot U_{dpe,rec} \cdot I_d - R_{loop} \cdot I_d^2. \quad (8)$$

Equation (8) is the load flow equation of the link. Everything in Section 4 follows from solving it. The factor k is the only formal difference between a link running on two poles and one running on one: with two poles energised the flow works against twice the pole-to-ground voltage, with one pole out against $U_{dpe,rec}$ alone, between the healthy pole and the return.

The loss splits between the conductors in proportion to their resistances, which is reported separately because the DMR contribution is what distinguishes the outage case:

$$P_{loss,HV} = \begin{cases} 2 \cdot R_{d,hv} \cdot I_d^2 & \text{balanced,} \\ R_{d,hv} \cdot I_d^2 & \text{one pole out,} \end{cases} \quad P_{loss,DMR} = \begin{cases} 0 & \text{balanced,} \\ R_{d,dmr} \cdot I_d^2 & \text{one pole out.} \end{cases} \quad (9)$$

The two contributions must add to $R_{loop} \cdot I_d^2$ and no more: in the one-pole-out case the HV term carries $R_{d,hv}$ alone, not the loop resistance, or the DMR loss is counted twice.

3.5 Receiving-end voltage

The voltage at the far end of the line is the sending-end voltage less the drop along the conductors the current traverses. With two poles energised the arrangement is symmetric — each conductor drops $I_d \cdot R_{d,hv}$ and the midpoint stays at ground — so the pole-to-ground voltage at the inverter is

$$U_{dpe,inv} = U_{dpe,rec} - I_d \cdot R_{d,hv}, \quad (10)$$

which is consistent with Equation (8), since $2 \cdot U_{dpe,inv} \cdot I_d = 2 \cdot U_{dpe,rec} \cdot I_d - 2 \cdot R_{d,hv} \cdot I_d^2$.

With one pole out, Equation (10) still gives the pole-to-ground voltage of the healthy pole at the inverter, but it is no longer the voltage across the converter: the DMR carries the return current and rises by $I_d \cdot R_{d,dmr}$ above the rectifier neutral, so the converter sees

$$U_{dpm,inv} = U_{dpe,rec} - I_d \cdot R_{loop} \quad (\text{one pole out}). \quad (11)$$

Both figures are useful — the first for insulation coordination against ground, the second for the converter operating point — and the distinction matters only in the outage case.

4 Solution of the load flow

4.1 Why the current is not explicit

If both the power and the voltage were specified at the same end of the line, the current would follow from a division. They are not: power is fixed at the receiving end and voltage at the sending end, and the two are separated by a loss that itself depends on the current. Substituting the target power into Equation (8) therefore gives a quadratic in I_d rather than a linear relation — one that can be solved in closed form, so no iteration is needed.

4.2 The current

Start from the DC power that must arrive at the inverter, Equation (6):

$$P_{dc,inv} = \frac{P_{ac,inv}}{1 - p_{inv}}. \quad (12)$$

Insert it into Equation (8) and collect terms:

$$R_{loop} \cdot I_d^2 - k \cdot U_{dpe,rec} \cdot I_d + P_{dc,inv} = 0. \quad (13)$$

The operating point is the **lower** root:

$$I_d = \frac{k \cdot U_{dpe,rec} - \sqrt{k^2 \cdot U_{dpe,rec}^2 - 4 \cdot R_{loop} \cdot P_{dc,inv}}}{2 \cdot R_{loop}} \quad (14)$$

with $U_{dpe,rec}$ in kV, R_{loop} in Ω , $P_{dc,inv}$ in MW and I_d in kA. Equation (14) covers both operating states through k . With two poles energised ($k = 2$) it simplifies to

$$I_d = \frac{U_{dpe,rec} - \sqrt{U_{dpe,rec}^2 - R_{loop} \cdot P_{dc,inv}}}{R_{loop}}. \quad (15)$$

Both roots of Equation (13) deliver the same power to the inverter, but they do it at different efficiencies. The upper root is the high-current solution on the far side of the line's maximum power point, where more current is pushed through the line only to burn the extra in $I^2 \cdot R$. It is not an operating point any converter control would settle at, and it lies far outside the current rating of the conductors. The lower root is the physical one.

4.3 Maximum transferable power

The square root in Equation (14) becomes imaginary when the requested power exceeds what the line can deliver at that voltage. The limit follows from $dP_{dc,inv}/dI_d = 0$ applied to Equation (8):

$$I_d|_{P_{max}} = \frac{k \cdot U_{dpe,rec}}{2 \cdot R_{loop}}, \quad P_{max} = \frac{k^2 \cdot U_{dpe,rec}^2}{4 \cdot R_{loop}}. \quad (16)$$

P_{max} is a bound on solvability, not a capability: at that point half the sending-end power is dissipated in the line. For any sensible design it sits well above the rating, and a real link is limited by conductor thermal rating and valve current long before it. Its use is as a sanity check — if a requested transfer has no solution, the combination of power, voltage and line resistance is the reason. The outage state has one quarter the maximum of balanced operation at the same voltage and, with the DMR in the loop, less than that again.

4.4 The rest of the flow

With the current known, the remaining quantities follow directly:

$$\begin{aligned}
 P_{dc,rec} &= k \cdot U_{dpe,rec} \cdot I_d & \text{Eq. (7)} & (17) \\
 P_{loss,line} &= R_{loop} \cdot I_d^2 & \text{Eq. (7), split by Eq. (9)} & (18) \\
 U_{dpe,inv} &= U_{dpe,rec} - I_d \cdot R_{d,hv} & \text{Eq. (10)} & (19) \\
 P_{ac,rec} &= \frac{P_{dc,rec}}{1 - p_{rec}} & \text{Eq. (5)} & (20) \\
 P_{loss,total} &= P_{ac,rec} - P_{ac,inv} & \text{AC-to-AC loss} & (21) \\
 \eta &= \frac{P_{ac,inv}}{P_{ac,rec}} & \text{overall efficiency} & (22)
 \end{aligned}$$

The three loss contributions account for the whole difference between the two ends:

$$P_{ac,rec} - P_{ac,inv} = \underbrace{p_{rec} \cdot P_{ac,rec}}_{\text{rectifier}} + \underbrace{R_{loop} \cdot I_d^2}_{\text{line}} + \underbrace{p_{inv} \cdot P_{dc,inv}}_{\text{inverter}} \quad (23)$$

4.5 Current margin

The margin does not enter the flow. It scales the operating current to a rating figure for conductor selection, valve and bushing ratings, switchgear and protection settings:

$$I_{rated} = (1 + m) \cdot I_d. \quad (24)$$

Losses, voltage drops and powers all belong to the operating current I_d , not to I_{rated} ; the margined value is a sizing quantity only. A worst-case loss figure comes from re-evaluating the flow at a higher power, not from substituting I_{rated} into the loss relation, which would be inconsistent with the stated transfer.

Because the outage state can carry the highest current of all the states considered, the governing rated current for the conductor and the valves is the largest I_{rated} across them, not the one belonging to normal operation.

Given	Unit	SMP	DSMP	BIP	BIP 1 pole
AC power at receiving end	MW	1,000	1,500	3,000	1,500
DC voltage pole-to-ground $U_{dpe,rec}$	kV	400	400	525	525
DC line length	mi	160	200	200	200
HV conductor	—	Bluebird	Chukar	Bluebird	Bluebird
HV subconductors per pole	—	1	2	2	2
Specific HV resistance	$\Omega/1000$ ft	0.00801	0.00970	0.00801	0.00801
DMR conductor	—	n/a	n/a	Chukar	Chukar
DMR subconductors	—	n/a	n/a	2	1
Specific DMR resistance	$\Omega/1000$ ft	n/a	n/a	0.00970	0.00970
Reference temperature T_{ref}	$^{\circ}\text{C}$	20	20	20	20
Conductor temperature T	$^{\circ}\text{C}$	75	75	75	75
Zero-resistance temperature T_0	$^{\circ}\text{C}$	228	228	228	228
Losses per converter station	%	1.2	1.2	1.2	1.2
Margin for DC current	%	5	5	5	5

Table 3: Given quantities for the four cases. BIP 1 pole is the 3 GW bipole with one pole out of service, on the same plant as the BIP case and with its own post-contingency transfer.

5 Worked cases

5.1 The cases

Four operating cases are worked through below: the 1 GW symmetrical monopole, a double symmetrical monopole variant, the 3 GW bipole, and that bipole holding half its transfer on one pole. Station losses are 1.2 % per station and the current margin 5 %. All conductor data is the DC resistance at 20 $^{\circ}\text{C}$ from the ACSR tables and the flow is evaluated at a conductor temperature of 75 $^{\circ}\text{C}$.

5.2 Temperature correction and line resistance

The correction factor is common to all four cases, Equation (2):

$$k_T = \frac{228 + 75}{228 + 20} = 1.2218.$$

Applying Equation (3) to the bipole, whose HV conductor and length the outage case shares:

$$R_{d,hv} = \frac{0.00801 \cdot 5.28}{2} \cdot 200 \cdot 1.2218 = 4.229 \cdot 1.2218 = 5.167 \, \Omega,$$

and to the single-subconductor DMR that carries the return in the outage case:

$$R_{d,dmr} = \frac{0.00970 \cdot 5.28}{1} \cdot 200 \cdot 1.2218 = 10.243 \cdot 1.2218 = 12.515 \, \Omega.$$

So $R_{loop} = 10.335 \, \Omega$ balanced and 17.682 Ω with the return through the DMR. At the 20 $^{\circ}\text{C}$ data temperature those figures would have been 8.459 Ω and 14.472 Ω : the correction alone adds 22 % to every resistive loss reported below.

5.3 Balanced bipole, step by step

Following Section 4 with $k = 2$:

$$P_{dc,inv} = \frac{3000}{1 - 0.012} = 3036.4 \text{ MW} \quad \text{Eq. (12)}$$

$$I_d = \frac{525 - \sqrt{525^2 - 10.335 \cdot 3036.4}}{10.335} = \frac{525 - 494.21}{10.335} = 2.979 \text{ kA} \quad \text{Eq. (15)}$$

$$P_{dc,rec} = 2 \cdot 525 \cdot 2.979 = 3128.2 \text{ MW} \quad \text{Eq. (17)}$$

$$P_{loss,line} = 10.335 \cdot 2.979^2 = 91.7 \text{ MW} \quad \text{Eq. (18)}$$

$$U_{dpe,inv} = 525 - 2.979 \cdot 5.167 = 509.6 \text{ kV} \quad \text{Eq. (10)}$$

$$P_{ac,rec} = \frac{3128.2}{1 - 0.012} = 3166.2 \text{ MW} \quad \text{Eq. (20)}$$

$$\eta = \frac{3000}{3166.2} = 94.75 \% \quad \text{Eq. (22)}$$

The loss balance of Equation (23) closes: $38.0 + 91.7 + 36.4 = 166.2 \text{ MW} = P_{ac,rec} - P_{ac,inv}$.

5.4 One pole out, step by step

The same plant with $k = 1$, the DMR in the loop, and its own 1500 MW target:

$$P_{dc,inv} = \frac{1500}{1 - 0.012} = 1518.2 \text{ MW}$$

$$I_d = \frac{525 - \sqrt{525^2 - 4 \cdot 17.682 \cdot 1518.2}}{2 \cdot 17.682} = \frac{525 - 410.16}{35.36} = 3.247 \text{ kA} \quad \text{Eq. (14)}$$

$$P_{dc,rec} = 525 \cdot 3.247 = 1704.6 \text{ MW}$$

$$P_{loss,line} = 5.167 \cdot 3.247^2 + 12.515 \cdot 3.247^2 = 54.5 + 131.9 = 186.4 \text{ MW} \quad \text{Eq. (9)}$$

Half the power, and yet a higher current and twice the line loss of the balanced bipole. This is the case that sizes the conductor.

5.5 Results

Table 4 collects the four cases, each at the 75°C conductor temperature.

5.6 Reading the results

Four points are worth drawing out.

The outage state sets the current rating. At 3247 A the bipole holding half its transfer on one pole draws 9% more current than it does at full power on two, and with the margin applied it asks for 3409 A against 3128 A in normal operation. A conductor or valve sized on balanced operation alone would be undersized for the contingency it exists to survive.

The DMR dominates the outage loss. 132 MW of the 186 MW lost in the one-pole-out case is in the DMR alone, because a single Chukar subconductor is 2.4 times the resistance of the two-conductor Bluebird pole bundle it stands in for. Doubling the DMR to two subconductors brings $R_{d,dmr}$ to 6.257 Ω, the current to 3101 A and the line loss to 110 MW — a 76 MW improvement for one extra conductor, and the first thing to settle in the DMR design.

Temperature is not a detail. Every resistance and every line loss in Table 4 is 22% higher than the conductor table suggests, purely because the line is evaluated hot. Carried through to the bipole efficiency the correction is worth about half a percentage point — the same order as the difference between the conductor options being compared, so a comparison made at data temperature would not be a fair one.

Result	Unit	SMP	DSMP	BIP	BIP 1 pole
Temperature correction factor	—	1.2218	1.2218	1.2218	1.2218
HV resistance per pole	Ω	8.268	6.257	5.167	5.167
DMR resistance	Ω	—	—	—	12.515
Loop resistance	Ω	16.535	12.515	10.335	17.682
DC power at inverter terminals	MW	1,012.1	1,518.2	3,036.4	1,518.2
DC current, operating	A	1,300	1,958	2,979	3,247
DC current, rated (5 % margin)	A	1,365	2,056	3,128	3,409
DC power at rectifier terminals	MW	1,040.1	1,566.2	3,128.2	1,704.6
Line losses, HV conductors	MW	27.9	48.0	91.7	54.5
Line losses, DMR	MW	—	—	0.0	131.9
Total DC line losses	MW	27.9	48.0	91.7	186.4
DC voltage at inverter $U_{dpe,inv}$	kV	389.3	387.7	509.6	508.2
DC voltage at inverter $U_{dpm,inv}$	kV	—	—	—	467.6
AC power at sending end	MW	1,052.7	1,585.2	3,166.2	1,725.3
Total losses, AC to AC	MW	52.7	85.2	166.2	225.3
Overall efficiency	%	94.99	94.62	94.75	86.94
Maximum transferable power	MW	9,676	12,785	26,671	3,897

Table 4: Results at a conductor temperature of 75 °C.

Where the power is defined changes the current. Taken as the power *leaving* the rectifier over a line at data temperature, 3000 MW on the bipole corresponds to $P/(2 \cdot U_{dpe,rec}) = 2857$ A. Required to *arrive* at the receiving AC terminals through a hot line, the same 3000 MW needs 2979 A, because the current now has to cover the line and converter losses as well. The second figure is the one to size equipment against when the transfer is guaranteed at the point of delivery.

Assumption	Consequence and justification
Purely resistive DC line	No inductance or capacitance appears in the relations. Correct for a steady-state DC flow; line charging and inductance matter only for transients and fault studies.
Uniform conductor temperature	The whole route sits at the single design temperature T . In service the temperature varies with ambient, wind and loading along the line, so the reported loss is that of the stated design condition rather than of any particular hour.
Linear resistance-temperature law	Equation (1) is the standard inferred-zero-temperature form, accurate to well within a percent from ambient to the maximum continuous rating of an ACSR conductor.
Aluminium carries the current	The steel core of an ACSR conductor is neglected in the resistance, which is standard practice; $T_0 = 228^\circ\text{C}$ is the aluminium constant.
Ideal voltage regulation at the rectifier	$U_{dpe,rec}$ holds exactly at the sending terminals irrespective of loading. Real control holds it within a droop band.
Station losses proportional to throughput	No no-load term, so light-load efficiency is overstated while full-load efficiency is representative. Adequate for screening; a two-term model ($p_0 + p_1 P$) is the natural refinement.
Balanced bipole carries no DMR current	The pole currents are taken as exactly equal. In service a small unbalance flows in the DMR; its loss contribution is second order, going with the square of a few percent of I_d .
Single operating point	Each case is one dispatch, not a duty cycle. Annual energy losses require a loading profile and a separate loss-energy calculation.

Table 5: Assumptions underlying the load flow relations.

6 Assumptions and limitations

6.1 Assumptions

6.2 Outside the scope

The following are deliberately not treated here, each belonging to a dedicated study:

- **Reactive power and AC-side voltage.** Only active power crosses the converter boundary in these relations. Converter reactive capability and AC network voltage support belong to the AC side of the design and, ultimately, to interconnection studies.
- **Ground and electrode return.** Only metallic return is treated. A bipole operating with earth return has a different loop resistance and a set of electrode, corrosion and interference constraints that sit outside a load flow.
- **Thermal rating of the conductor.** The flow yields the current and assumes a conductor temperature; whether that current actually produces that temperature at the ambient and wind conditions of the route is an ampacity calculation, and the two should be reconciled once the route conditions are fixed.
- **Converter overload and short-term ratings.** The margin of Equation (24) is a continuous rating margin, not an overload capability.
- **Multi-terminal and intermediate taps.** The relations assume a two-terminal, point-to-point link.
- **Dynamics.** Control response, ramp rates, pole-block transients and the transition into the one-pole-out state are all outside a steady-state flow.

6.3 Open items

- (1) **DMR conductor sizing.** The balanced bipole is described with a two-subconductor Chukar DMR and the outage case with one. Since the DMR only carries current when a pole is lost, the outage figure is

the one that matters, and the difference is 76 MW of line loss (Section 5). To be settled with the real DMR design.

- (2) **Conductor temperature basis.** 75 °C is the maximum continuous temperature and gives a worst-case loss and voltage drop. Whether losses should also be reported at an average operating temperature is a reporting decision still open — the two differ by of the order of 10 % in loss.
- (3) **Converter loss figure.** 1.2 % per station is a generic VSC-class value. To be replaced with vendor loss curves, at which point the two-term loss model of Table 5 becomes worthwhile.
- (4) **Loss-energy evaluation.** A loading profile and an annual loss-energy figure are the natural extension, since it is energy rather than peak loss that enters an economic comparison of voltage levels and conductor sizes.

7 References

- [1] Southwire Company, *ACSR — Aluminum Conductor, Steel Reinforced, Bare*, catalog section 11-4, 4 pp. Filed with this report as `references/ACSR-Southwire-11-4.pdf`. The two resistance columns are headed *DC @ 20°C* and *AC @ 75°C*, both in $\Omega/1000$ ft.